October 2026

ChemRxiv Editorial Team

Re: Submission of "Beyond Candidate Generation: A Validation-First Protocol for AI-Assisted Transition-State Discovery"

Dear Editors,

I am pleased to submit the manuscript above for consideration as a ChemRxiv preprint. The manuscript is a methods and perspective contribution on how to validate transition-state (TS) candidates produced by AI models. It is not a claim of a new predictive model; it is a protocol that separates candidate generation from physical validation and makes the validation layer reproducible.

The central argument is that a generated geometry is a hypothesis, not a transition state. A TS requires one imaginary frequency, the intended atom pair, the correct connectivity, and a consistent electronic state. The manuscript presents a four-stage protocol—chemical preflight, barrier-region scan, constrained reference plus Hessian, and validation—and implements it as a portable, open-source skill package for local ORCA/xtb and Gaussian/Slurm execution. It illustrates the protocol with internal case studies from Ni/Fe catalysis, including a per-point scan strategy that completed 16/16 points after a collective scan failed three times, and a generative-model test on an 84-atom broken-symmetry organometallic system where the candidates were out of distribution.

I believe the work is well suited to ChemRxiv because it is a community-facing methods contribution: the protocol and the implementation are meant to be used and criticized before journal submission. The preprint will also establish a citable reference for the open-source package. The manuscript is not under consideration elsewhere, and I am the sole author.

The author declares no competing financial interest. The code and manuscript source are available at `https://github.com/lhfjjk/ts-computation-suite`.

Suggested categories: Computational Chemistry (primary); Catalysis (secondary).

Sincerely,

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